

Zhuoning (Johnny) XU

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EDUCATION

New York University

New York, NY

MS, Computer Science (Courant Institute of Mathematical Sciences)

Sept. 2025 - May 2027 (expected)

- **Coursework:** Programming Language (Racket), Fundamental Algorithms, Mathematical Techniques for CS Applications, Machine Learning (Spring 2026), Operating Systems (Spring 2026), Natural Language Processing (Spring 2026)

The Hong Kong Polytechnic University

Hong Kong

BEng, Major in Electronic & Information Engineering; Minor in Computing (GPA: 3.8, Top 5%)

Sept. 2021 - Jun. 2025

- **Honors:** Dean's Honors List (2021-2024), First-class honours degree
- **Coursework:** Computer System Fundamentals (**x86 Assembly**), Programming Fundamentals (C/C++), Computer Programming (**Python**), Object-Oriented Programming (**Java**), Web Application Design and Development (**Flask**), Operating Systems, Database System (**SQL&NoSQL**), Data and Computer Communication, Data Structures, Computer Vision and Pattern Recognition, Mobile APP Programming (**Android**), Service and Cloud Computing (**RESTful API, AWS Cloud Computing**), Computer Systems Security

SKILLS

- **Programming Languages:** Assembly Language, C/C++, **Java**, Web programming (PHP, HTML, CSS, JavaScript, AJAX), **Python** (Pandas, NumPy, Scikit-Learn, Matplotlib, **PyTorch**), SQL
- **Development Tools:** MySQL Workbench, Matlab, Linux Shell, PyCharm, Docker, Android Studio, MS Office Suite, Git

INTERNSHIP EXPERIENCES

Assistant Software Engineer Intern - Front-end & Data Monitoring, O&M Software Department

Shenzhen, China

YOTAI Digital Energy Technology Co., Ltd

Jun. 2024 - Aug. 2024

- Resolved data loss, errors, and UI lag by refining Umi/**Node.js** and shipping reusable hooks (**WebSocket, device model, device data**); improved page responsiveness by 15%
- Built the monitoring web app and a unified equipment model, standardizing backend APIs and stabilizing live views with an efficient subscription and update pipeline
- Developed a **Python analysis module** to automate reporting, boosting metric accuracy, and cutting preparation time

Data Engineer Intern - Business Intelligence & Visualization, Technical Department

Hangzhou, China

Hangzhou Green Cloud S & T Co., Ltd.

Jun. 2023 - Jul. 2023

- Used **SQL** to collect, clean, and combine operational data so dashboards had trustworthy inputs and analysts spent less time on one-off data requests.
- Led desktop and **mobile report development** for 10+ clients; built **data visualization** platforms with FineBI and launched command-center dashboards and large-screen displays with clear, interactive designs.
- Captured requirements and streamlined processes to design custom reports; applied **statistical analysis** to deliver forecasts and recommendations that improved operational efficiency

RESEARCH EXPERIENCES

A Multi-Perceptual Learning Network for Retina OCT Image Denoising and Classification

Jan. 2024 - Sept. 2024

- Replaced VGG with **ResNet** and integrated **Multi-Perception Learning** with a Triple-Cross-Fusion GAN (**PyTorch/OpenCV**) for joint denoising and classification
- **Improves image quality** and **increases classification accuracy** by over 1% compared with the baseline on the same dataset, while maintaining clinician-friendly real-time inference on GPU

A Sensor-based System for Monitoring and Providing Feedback on Physical Activity in Older Adults

Sept. 2023 - Sept. 2024

- Built an end-to-end pipeline with **YOLOv8-Pose**, detecting 17 keypoints per frame and joint angles converted into sequences for an **LSTM classifier**
- Evaluated on 200 samples with a confusion matrix showing overall accuracy above 90 percent

Transformer-based Vision-Language Jigsaw Puzzle Assembly (Accepted by AAAI-26 Student Abstract Program)

Sept. 2025

- Built a network using **vision transformer** backbones and vision-language model to align fragment patches with text at different levels; implemented Multi-Head Attention Fusion and permutation assignment in PyTorch
- Drove full ML lifecycle: modular data/preprocessing, reproducible training, experiment tracking & ablations, and model packaging for efficient inference, showcasing strengths in multimodal modeling and systems engineering

PROJECT EXPERIENCES

Robotics Software Development - Software Development for Automated Cart Delivery

Sept. 2023 - May 2024

- Implemented multi-sensor fusion on **ROS** (motor encoders, IR, etc.) with closed-loop pose control via **PID**; built **Python** and C tooling for **real-time sensor analysis** and robotic-arm positioning
- Achieved stable route following with quick disturbance recovery; end-to-end travel-distance error within **5%**, with turns handled without overshoot/oscillation

Deep Learning for Advanced Driver Assistance - Traffic Sign Detection

Feb. 2024 - Feb. 2025

- Built a two-stage pipeline to improve traffic-sign recognition in adverse weather by pairing a **neural-network-based** image restoration module with a classifier
- Achieved 25 dB PSNR on restored images and about 80 percent classification accuracy, indicating practical potential for real-world deployment

PUBLICATIONS

- "A Multi-Perceptual Learning Network for Retina OCT Image Denoising and Classification" 2024 Asia Pacific Signal and Information Processing Association Annual Summit and Conference (DOI: 10.1109/apsipaasc63619.2025.10848686)
- "MFGAN: OCT image super-resolution and enhancement with blind degradation and multi-frame fusion" International Workshop on Advanced Imaging Technology (IWAIT) 2025 (DOI: 10.1117/12.3057230)